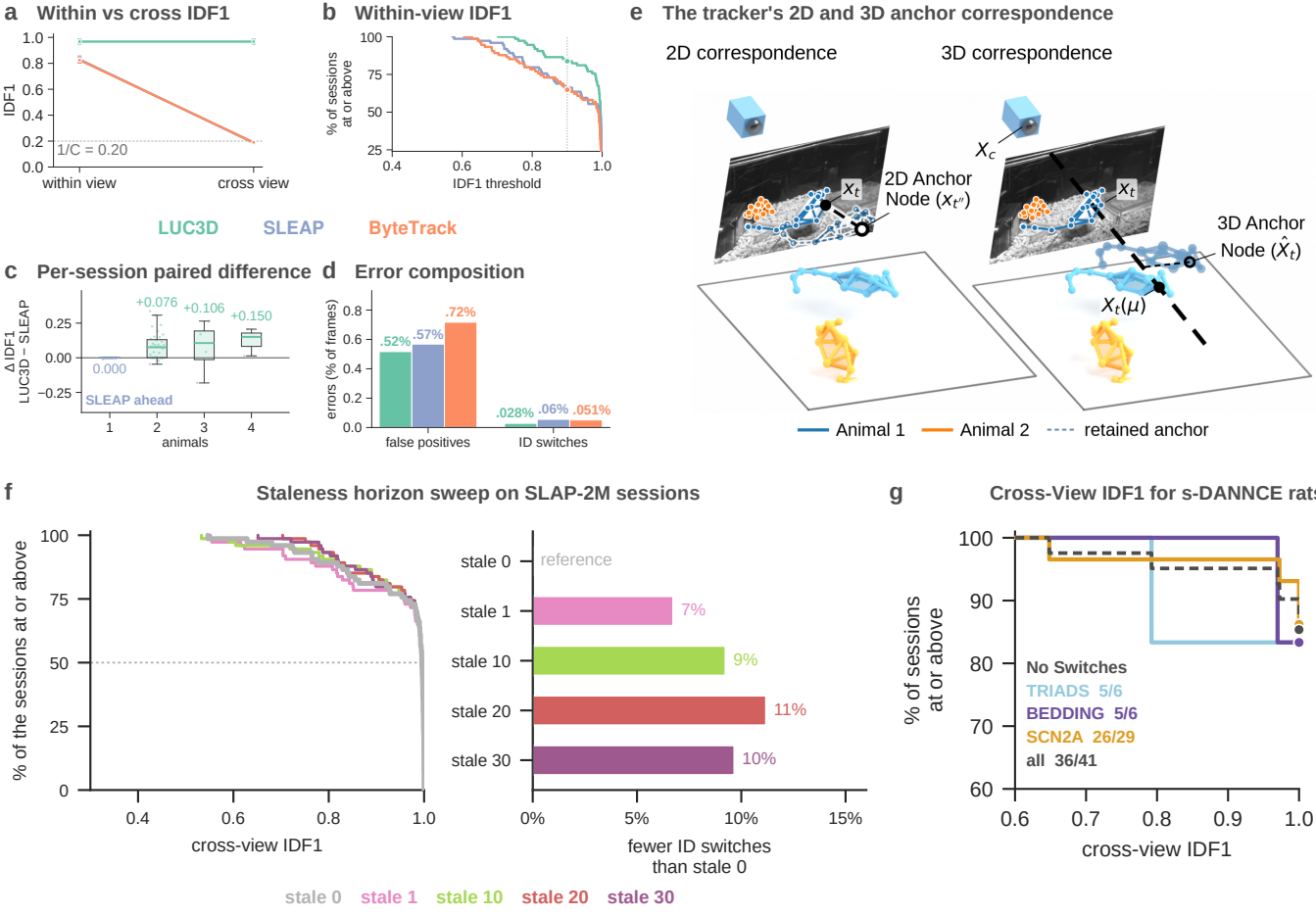

a
Within vs cross IDF1
IDF1
1.0
0.8
0.6
0.4
0.2
0.0
1/C = 0.20
within view
cross view
b
Within-view IDF1
% of sessions at or above
100
75
50
25
0.4
0.6
0.8
1.0
IDF1 threshold
LUC3D
SLEAP
ByteTrack
c
Per-session paired difference
ΔIDF1 LUC3D − SLEAP
0.25
0.00
−0.25
+0.076
+0.106
+0.150
0.000
SLEAP ahead
1
2
3
4
animals
d
Error composition
errors (% of frames)
0.8
0.6
0.4
0.2
0.0
.52%
.57%
.72%
.028%
.06%
.051%
false positives
ID switches
e
The tracker's 2D and 3D anchor correspondence
2D correspondence
3D correspondence
x_t
2D Anchor Node ($x_{t'}$)
X_c
x_t
3D Anchor Node ($\hat{X}_t$)
$X_t(\mu)$
Animal 1
Animal 2
retained anchor
f
Staleness horizon sweep on SLAP-2M sessions
% of the sessions at or above
100
75
50
25
0
0.4
0.6
0.8
1.0
cross-view IDF1
stale 0
stale 1
stale 10
stale 20
stale 30
reference
7%
9%
11%
10%
0%
5%
10%
15%
fewer ID switches than stale 0
stale 0
stale 1
stale 10
stale 20
stale 30
g
Cross-View IDF1 for s-DANNCE rats
% of sessions at or above
100
90
80
70
60
0.6
0.7
0.8
0.9
1.0
cross-view IDF1
No Switches
TRIADS 5/6
BEDDING 5/6
SCN2A 26/29
all 36/41